Hybrid Semantic Context Forests and Kyphosis-Aware Deformable Topology Matching for Automatic 3D Rib Centerline Extraction and Labeling in Thoracic CT

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Abstract. Automatic extraction and anatomical numbering (`R01`–`R12`, `L01`–`L12`) of all 24 rib centerlines in thoracic CT is a prerequisite for unfolded rib visualisation, fracture detection and surgical navigation, but remains difficult because of adjacent-bone confluence (clavicle, scapula, sternum, vertebrae), calcified costal cartilage bridges, kyphotic pose variation and the frequent partial visibility of the first and twelfth ribs. We present `ribline`, an open-source three-stage framework that combines (i) an automatically detected lung-apex / vertebral-midline coordinate frame with a kyphosis-aware azimuthal angle used to order points along each rib, (ii) a hybrid voxel detector that max-fuses an anatomically anchored Semantic Context Forest (SCF) with a lightweight 3D U-Net, and (iii) a global labeling stage that decomposes the probability skeleton into candidate curves, assigns rib indices with a patient-scaled bilateral Viterbi dynamic program coupled across the two hemithoraces, and refines terminal ribs with template recovery and normal-plane centring (`RibSnake`). On the 60-scan Siemens Corporate Research (SCR) 2 mm thoracic CT benchmark (47 training / 12 evaluation scans), using a symmetric 10 mm labeling criterion and a 50 mm penalty for missed ribs, the full pipeline reaches a labeled symmetric Chamfer distance of 2.05 ± 0.48 mm (95 % bootstrap CI $[1.80, 2.33]$), labeled clDice of 0.886 (3 mm) / 0.945 (4 mm), 97.2 % labeling accuracy (278/286 ribs, Wilson 95 % CI 94.6–98.6 %), 97.2 % recall and no false-positive curves. We report every metric with confidence intervals, make the metric definitions, training recipes and checkpoints consistent, and state explicitly that the Stage-3 constants were developed with access to the 12 evaluation scans. Applied unchanged to the 47 training scans (1,116 ribs), on which those constants were never inspected, the frozen pipeline reaches 2.63 ± 1.91 mm (median 1.89 mm) and 95.4 % labeling accuracy; in a 5-fold cross-validation over all 59 scans (1,402 ribs) with per-fold refitted atlas, SCF and 3D U-Net and frozen Stage-3 constants, it reaches 3.64 ± 3.90 mm (95 % CI $[2.79, 4.74]$, median 2.39 mm) and 91.8 % labeling accuracy ($p < 10^{-5}$ over standalone detectors and $p = 2 \times 10^{-9}$ for `RibSnake` refinement).

Code: https://github.com/wq2012/ribline
Pretrained Models: https://huggingface.co/collections/wq2012/ribline

Keywords: Rib centerline extraction, anatomical rib labeling, semantic context forests, 3D U-Net, Viterbi dynamic programming, deformable template matching, thoracic CT

1 Introduction

The human rib cage comprises twelve pairs of curved, tubular bones (`R01`–`R12`, `L01`–`L12`). Accurate 3D extraction and individual numbering of rib centerlines in CT enables unfolded rib cage visualisation via curved planar reformation for trauma screening [9,10], provides an anatomical coordinate system for localising thoracic lesions [5], supports radiotherapy and spinal instrumentation planning [6], and is the topological backbone of automated rib-fracture detection [4].

Despite the high attenuation of cortical bone, fully automatic 3D rib centerline tracing and numbering is hard for four reasons: (1) *adjacent bone confluence*: the first rib articulates with the manubrium and lies next to the clavicle and scapula, so skeletonisation leaks into the shoulder girdle; (2) *intercostal ossification*: calcified costal cartilage and partial-volume averaging create bridges between ribs 5–10, producing false junctions and hairpin U-turns; (3) *terminal-rib variability*: the floating twelfth ribs vary widely in length and are congenitally absent in a few percent of patients, so naive top-down or bottom-up counting suffers index shifts whenever a terminal rib is missed; and (4) *thoracic kyphosis*: in the upper thorax the rib cage curves anteriorly and inferiorly while the spine arches posteriorly, so a polar angle around the torso centroid is not a reliable ordering parameter along the posterior rib neck.

Early methods relied on seeded region growing or ridge tracking [8,5,6,7]. Wu *et al.* [1] introduced learning-based deformable template matching with Marginal Space Learning [12] on the SCR data set used here, reporting 89.5–95.9 % labeling accuracy under their own protocol. Auto-context [13] and Semantic Context Forests (SCF) [2,3] showed that combining multi-scale differential features with long-range random-shift intensity/probability differences gives discriminative voxel classifiers for thin structures. Deep 3D networks [11,10] and point-cloud networks such as RibSeg v2 [4] achieve strong segmentation, but on small training cohorts standalone CNNs still under-segment the small first and floating twelfth ribs and need post-hoc heuristics for numbering.

Contributions. We describe and release (Fig. 1) a three-stage pipeline whose methods section below is written to match the released code line by line:

1. a **lung-apex / vertebral-midline thoracic frame** and a **kyphosis-aware azimuthal angle** $\theta(\mathbf{x})$ whose focus moves posteriorly towards the apex; on the 1,402 annotated ribs of the 59 valid SCR scans θ increases from the posterior to the anterior end of *every* rib and is strictly monotonic on 78.6 % of ribs (vs. 57.0 % for a torso-centre angle; Sec. 2);
2. a **complementary hybrid detector**: an anatomically anchored SCF (better on first ribs) max-fused with a plain 3D U-Net (better on intermediate ribs);

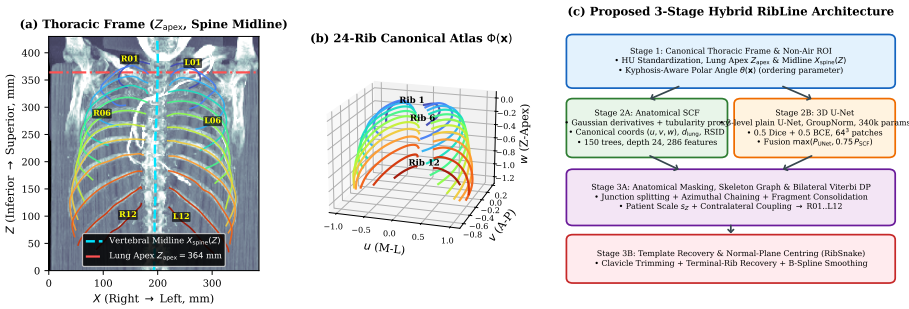


Fig. 1. Overview of `ribline`. (a) Coronal MIP with the detected lung apex Z_{apex}, the vertebral midline $X_{\text{spine}}(Z)$ and the ground-truth centerlines. (b) The 24-rib canonical atlas $\bar{\mathcal{C}}_{s,k}$ fitted on the 47 training scans. (c) Flowchart of the hybrid SCF + 3D U-Net detector, the bilateral Viterbi labeler and the `RibSnake` refinement.

3. a **patient-scaled bilateral Viterbi labeler** with contralateral coupling and outlier tolerance, plus `RibSnake` (clavicle trimming, terminal-rib template recovery, normal-plane centring);
4. a **corrected evaluation protocol** (symmetric labeling criterion, missed-rib penalty, precision, bootstrap and Wilson confidence intervals, paired tests), a transparent treatment of one annotation error in the public ground truth, and a cross-validation with frozen Stage-3 constants that quantifies how much the development protocol favoured the 12 evaluation scans.

2 Canonical Thoracic Frame and Statistical Rib Atlas

2.1 Hounsfield Standardisation and Thoracic Landmark Detection

The SCR archive mixes signed Hounsfield units (air ≈ -1000) and unsigned shifted values (air ≈ 0). If the minimum intensity of a volume is ≥ -100, we subtract 1024; otherwise the volume is used as is (`standardize_hu`).

All landmarks are computed on a 2× down-sampled grid. The torso mask is $I > -350\,\text{HU}$ with slice-wise hole filling; internal air is the filled torso minus voxels with $I \geq -380\,\text{HU}$, morphologically opened and reduced to the (at most four) largest components with more than 800 sub-voxels, which form the lung mask $\mathcal{M}_{\text{lung}}$. The **lung apex** Z_{apex} is the larger of the per-side 99.5th-percentile Z coordinates of $\mathcal{M}_{\text{lung}}$. The **spine** is located as the peak of a Gaussian-smoothed ($\sigma = 3$ sub-voxels) axial projection of bone intensity ($\text{clip}(I - 120, 0, 600)$) restricted to the posterior 40 % of the antero-posterior bone extent and to $\pm 50\,\text{mm}$ around the medio-lateral bone centre, refined by an intensity-weighted centroid; a per-slice median of bone voxels within $\pm 32 \times \pm 36\,\text{mm}$ of this seed, smoothed with a Gaussian of $\sigma = 12$ slices, yields the trajectory $(X_{\text{spine}}(Z), Y_{\text{spine}}(Z))$; the scalar seed values $(x_{\text{spine}}, y_{\text{spine}})$ are also kept. The thoracic half-width W is the 96th percentile of $|X - x_{\text{spine}}|$ and $Y_{\text{ant}}, Y_{\text{post}}$ the 5th/95th percentiles of Y

over bone voxels ($I > 150\,\mathrm{HU}$) in the band $Z_{\mathrm{apex}} - 210 \le Z \le Z_{\mathrm{apex}} - 10\,\mathrm{mm}$; $D = \max(Y_{\mathrm{post}} - Y_{\mathrm{ant}}, 100\,\mathrm{mm})$.

Measured on the 59 valid scans (script `polar_angle_stats`), the posterior (spine-side) end of `R01`/`L01` lies $+11.9 \pm 9.3\,\mathrm{mm}$ above Z_{apex} ($n = 117$; median 13.0 mm, 5–95 % range -2.5 to 24.5 mm), and the midline error $|X_{\mathrm{spine}}(Z) - \frac{1}{2}(X^{\mathrm{R}}_{\mathrm{post}} + X^{\mathrm{L}}_{\mathrm{post}})|$ over 695 annotated rib pairs is $2.5 \pm 2.4\,\mathrm{mm}$ (median 1.8 mm). The apex is therefore a useful but not a tight anchor for the first rib; the labeler below never relies on it alone.

2.2 Canonical Mapping and Kyphosis-Aware Azimuthal Angle

A physical point $\mathbf{x} = (X, Y, Z)$ is mapped to canonical coordinates

$$u = \frac{X - x_{\mathrm{spine}}}{W}, \qquad v = \frac{Y - y_{\mathrm{spine}}}{D}, \qquad w = \frac{Z - Z_{\mathrm{apex}}}{240\ \mathrm{mm}}, \tag{1}$$

using the scalar spine seed (the per-slice trajectory is used only for the anatomical masks of Sec. 4). To order points along a rib from its vertebral head to its sternal end we use an azimuthal angle whose focus sits $x_c = 8\,\mathrm{mm}$ lateral of the midline and moves posteriorly towards the apex,

$$y_c(Z) = Y_{\mathrm{ant}} + 0.25\,D + 0.16\,\mathrm{clip}(Z_{\mathrm{apex}} - Z,\ 0,\ 220\ \mathrm{mm}), \tag{2}$$

$$\theta(\mathbf{x}) = \mathrm{atan2}\big(-(Y - y_c(Z)),\ |X - x_{\mathrm{spine}}| - x_c\big), \tag{3}$$

followed by phase unwrapping along the curve. Over the 1,402 ground-truth ribs, $\theta_{\mathrm{end}} > \theta_{\mathrm{start}}$ (correct global orientation) holds for 100 % of ribs; θ is *strictly* monotonic for 1,102 ribs (78.6 %), the largest local reversal exceeds 0.1 rad on 8 ribs (0.6 %) and 0.3 rad on 3 ribs. The naive angle $\mathrm{atan2}(-(Y - y_{\mathrm{lung}}), |X - x_{\mathrm{spine}}|)$ around a fixed torso centre is strictly monotonic for only 799 ribs (57.0 %) with reversals > 0.1 rad on 107 ribs (7.6 %). All downstream uses of θ (orientation, fragment chaining, angular span) are robust to small reversals.

2.3 Non-Parametric 24-Rib Statistical Atlas

From the 47 training scans (1,116 ribs) each rib is oriented posterior-to-anterior, resampled to $M = 80$ arc-length-uniform points, mapped by Eq. 1 and averaged to a mean curve $\bar{\mathcal{C}}_{s,k} \in \mathbb{R}^{80\times 3}$ per side s and index k (Fig. 1b). For every k we also store mean and standard deviation $(\boldsymbol{\mu}_{f,k}, \boldsymbol{\sigma}_{f,k})$ of the 5-D descriptor $\mathbf{f}(\mathcal{C}) = [z_{\mathrm{post}},\ \bar{z},\ x_{\max}/W,\ L/W,\ y_{\min}/D]$ (posterior-end height and mean height relative to Z_{apex}, normalised lateral reach, normalised arc length, normalised most-anterior Y), the inter-rib spacing $(\mu_{s,k}, \sigma_{s,k})$ between the posterior ends of ribs k and $k+1$, and a 15,000-point cloud of canonical points with their height offset from the rib's posterior end. The latter provides a *slant-corrected posterior-equivalent height* z_{sort} for any curve or fragment by 16-nearest-neighbour regression, which is the quantity used to order candidates craniocaudally. We verified that re-fitting the atlas with the released `build_atlas` script on the 47 training scans reproduces the shipped atlas exactly (maximum absolute difference 0), and that fitting on all 59 scans does not.

3 Hybrid Semantic Context Forest and 3D U-Net

3.1 Semantic Context Forests

Following [2], voxel-wise centerline detection is cast as classification. For every voxel we extract the raw intensity and, at $\sigma \in \{1, 2, 4\}$ mm, the Gaussian-smoothed intensity, gradient magnitude, the six second-order Gaussian derivatives and the Laplacian (28 features), plus Random Shift Intensity Differences (RSID)

$$f_{\mathrm{RSID}}(\mathbf{x}; \mathbf{u}_1, \mathbf{u}_2) = I(\mathbf{x} + \mathbf{u}_1) - I(\mathbf{x} + \mathbf{u}_2), \tag{4}$$

with integer voxel offsets drawn uniformly from $[-r, r]^3$. Three forests are released (scikit-learn random forests, balanced class weights, $\sqrt{d}$ features per split): **Pass-1** (120 trees, unlimited depth, 228 features): 200 RSID pairs with $r = 20$ voxels (± 40 mm). **Pass-2 auto-context** (100 trees, 306 features): the Pass-1 features, 28 multi-scale features of the Pass-1 probability map $P^{(1)}$ and 50 Random Shift Probability Differences $P^{(1)}(\mathbf{x} + \mathbf{v}_1) - P^{(1)}(\mathbf{x} + \mathbf{v}_2)$ with $r = 40$ voxels (± 80 mm) [13]. **Anatomically anchored SCF** (150 trees, depth 24, 286 features): Hounsfield-standardised input, 250 RSID pairs with $r = 12$ voxels (± 24 mm), a per-scale tubularity proxy built from the diagonal second derivatives, $\tau_\sigma = (-\min(0, I_{xx}) - \min(0, I_{yy}) - \min(0, I_{zz}))\,\rho/(\rho + 50)$ with $\rho = \|(I_{xx}, I_{yy}, I_{zz})\|$ (no eigen-decomposition or Frangi filter is computed), and five anatomical channels $|u|, u, v, w$ and the distance to the lung mask. Training samples 50,000 voxels per scan, half on the annotated centerlines and half uniformly over the whole volume; the Pass-1 forest additionally used random rigid augmentation of the training scan. At inference the forests are evaluated only on voxels with $G_{1.2}(I) > -20$ HU, i.e. everything except air; this ROI is a speed optimisation and is not used during training.

3.2 3D U-Net and Hybrid Fusion

The second detector is a plain three-level 3D U-Net ($16 \to 32 \to 64$ channels, two 3^3 convolutions with GroupNorm and ReLU per level, max-pooling and transposed-convolution up-sampling, no residual connections; 339,889 parameters). It is trained on 64^3 patches (70 % centred near a centerline voxel, 30 % uniformly), input normalised by a per-volume z-score of the *raw* stored intensities (no Hounsfield standardisation), target = centerline voxels dilated to a tube, loss = $0.5\,(1-\text{soft Dice}) + 0.5\,\mathrm{BCE}$, Adam ($10^{-3}$), batch 2, 8 patches per scan and epoch, 20 epochs, no augmentation. The released checkpoint `rib_unet_3d.pth` is the epoch-20 weights; it reproduces the probability maps used for all results below bit-exactly (maximum absolute difference 0 on re-inference). Inference uses 64^3 sliding windows with stride 48 and averaging.

The two detectors are complementary (Fig. 2): the SCF, with explicit canonical coordinates, is more sensitive on the short apical first ribs, whereas the U-Net gives cleaner shafts on intermediate ribs. We fuse the probability volumes by a voxel-wise maximum,

$$P_{\mathrm{hyb}}(\mathbf{x}) = \max(P_{\mathrm{UNet}}(\mathbf{x}),\ 0.75\,P_{\mathrm{SCF}}(\mathbf{x}))\,. \tag{5}$$

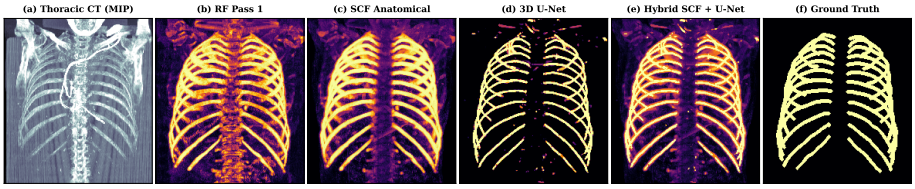


Fig. 2. Coronal MIPs of centerline probability maps on evaluation scan `pet_ct_bc__eh_Anonymous_0004`: (a) CT, (b) Pass-1 forest, (c) anatomically anchored SCF, (d) 3D U-Net, (e) hybrid max-fusion, (f) dilated ground truth.

4 Topology Matching and Deformable Refinement (Stage 3)

4.1 Anatomical Masking, Skeletonisation and Fragment Chaining

P_{hyb} is multiplied by an anatomical mask that keeps voxels within 3.5 mm of bone ($I > 115$ HU or $G_{1.2}(I) > 75$ HU), inside $[Z_{\text{apex}} - 365, Z_{\text{apex}} + 32]$ mm, outside the vertebral corridor ($|X - X_{\text{spine}}(Z)| < 10.5$ mm posteriorly) and sternal corridor (< 31 mm anteriorly), and, in the upper thorax ($Z > Z_{\text{apex}} - 140$ mm), closer than 26 mm to the lung (removing scapula and lateral clavicle). Vertical intercostal valleys ($G_{1.2}(I)$ at least 12 HU below both axial neighbours and $I < 135$ HU) are suppressed. After a light Gaussian ($\sigma = 0.55$ voxels) the map is thresholded at 0.45 and skeletonised [14]. The 26-connected skeleton graph is pruned of spurs shorter than 8 mm, junction voxels are removed to obtain elementary chains, short steep vertical bridges (< 28 mm with $|\Delta Z|/L > 0.72$) are discarded, chains are split at hairpin turns of the smoothed θ profile and at jumps of the slant-corrected height larger than 16 mm, and collinear fragments on the same hemithorax are chained posterior-to-anterior when their gap is < 44 mm, their heights agree within 12 mm, and the end tangents are aligned with the chord ($\cos > 0.62$; > 0.72 for touching fragments). Curves shorter than 22 mm are dropped.

4.2 Candidate Consolidation and Patient Scale

Per hemithorax (side by the mean X relative to x_{spine}), candidates are oriented by θ, described by $(\mathbf{f}, z_{\text{sort}}, \theta_{\min}, \theta_{\max}, L)$, and consolidated: fragments that are disjoint in θ, within 11.5 mm in z_{sort} and < 75 mm apart are bridged (greedy, closest first); upward-sloping anterior fragments ($L < 140$ mm, start > 30 mm from the midline, rising > 3 mm) and short lateral fragments ($L < 85$ mm starting > 66 mm from the midline or with $\theta_{\min} > 0.25$) are removed when at least six candidates remain; and of two candidates within 8.5 mm in z_{sort} whose mean curve distance is < 6.5 mm (or that are within 5.5 mm and very unequal in length) only the longer is kept. A patient scale $s_Z \in [0.80, 1.25]$ is the blend $0.45\, s_{\text{step}} + 0.55\, s_{\text{span}}$ of the median adjacent step of long (≥ 50 mm) candidates divided by 21.8 mm and the ratio of the observed to the atlas craniocaudal span.

4.3 Bilateral Viterbi Labeling

For candidate i and label $k \in \{1, \dots, 12\}$ the unary cost is

$$\tilde{U}(i,k) = \tfrac{1}{2} \sum_{d=1}^{5} w_d \Big(\tfrac{\hat{f}_{i,d} - \mu_{f,k,d}}{\sigma_{f,k,d}} \Big)^2 + 10\, \bar{d}\big(\Phi_{s_Z}(\mathcal{C}_i), \bar{\mathcal{C}}_{s,k}\big), \tag{6}$$

where the height features are rescaled by s_Z, $\bar{d}$ is the mean one-directional distance from 40 resampled canonical points to the template, and $\mathbf{w} = (1.1, 0.7, 1.15, 1.05, 0.45)$, or $(1.2, 0.6, 0.3, 0.2, 0.25)$ for partial candidates (angular span $< 1.35\,\mathrm{rad}$, $k \notin \{1, 12\}$). Costs are made relative, $U(i,k) = \tilde{U}(i,k) - \min_m \tilde{U}(i,m)$, with a re-added offset $\min_m \tilde{U} - 8$ for short candidates ($L < 45\,\mathrm{mm}$) so that only genuinely atypical fragments can become clutter. Left/right candidates whose z_{sort} differ by $< 7.5\,\mathrm{mm}$ have their unaries blended with weight $0.45 \exp(-\Delta z^2/32)$.

Candidates sorted by decreasing z_{sort} are labeled by a Viterbi recursion over strictly increasing k with at most two skipped labels ($\Delta k \le 3$). Skipping candidate j as clutter costs $\kappa_j = 65$ if $L_j \ge 115\,\mathrm{mm}$, 22 if $L_j \ge 55\,\mathrm{mm}$ and $\theta_{\min,j} < -0.45\,\mathrm{rad}$, and $2.5 + 8\,\mathrm{clip}((L_j - 25)/50, 0, 1)$ otherwise; transitions require $\Delta z \ge 4.5\,\mathrm{mm}$ and cost $(\Delta z - s_Z \mu_{k' \to k})^2 / (2(s_Z^2 \sigma_{k' \to k}^2 + 6))$ plus $2.5(\Delta k - 1)$ if $\Delta z \ge 1.55\, s_Z \mu_{s,k'}$ and $18(\Delta k - 1)$ otherwise; starting at label k costs $3(k-1)$ and ending at k costs $1.5(12 - k)$. After a first solution per side, a second pass subtracts $3 \exp(-\Delta z^2/32)$ from $U(i,k)$ whenever the contralateral rib labeled k lies within 8.5 mm, and the recursion is re-solved.

4.4 `RibSnake` Refinement

(1) *Apical trimming*: an `R01`/`L01` longer than 135 mm is cut where $|X - x_{\mathrm{spine}}|$ first exceeds $0.55\,W$ (if the remainder is $\ge 45\,\mathrm{mm}$) and capped at 135 mm arc length. (2) *Terminal-rib recovery*: a missing twelfth rib is searched in a box 22–44 mm below the posterior end of rib 11 for voxels with $P \ge 0.30$ and $\max_{3^3} I \ge 150\,\mathrm{HU}$ at least 18 mm from rib 11, accepted if ≥ 12 voxels with $\max P \ge 0.45$ exist and the contralateral twelfth rib is present or also recovered; a missing first or twelfth rib is otherwise synthesised from the patient-conditioned template (atlas curve displaced by a smoothed interpolation of the displacements observed on the detected ribs, with mirrored contralateral coupling), anchored in Z to the contralateral rib, kept only if $\ge 11\,\mathrm{mm}$ from the adjacent rib, and snapped point-wise to the probability-weighted bone centroid in a 7^3-voxel box when the local score $P\,\mathrm{clip}((G(I) - 60)/300, 0, 1.5)$ reaches 0.14. (3) *Normal-plane centring*: every curve is resampled to $\max(15, \lfloor L/2\,\mathrm{mm} \rceil)$ points; for 5 iterations each point moves by $0.35\times$ the offset of the intensity-and-probability-weighted centroid sampled on a 5×5 grid (spacing 1.25 mm, radius $\le \sqrt{6.5}\,\mathrm{mm}$) in the local normal plane, is smoothed along the curve ($\sigma = 1.2$ points, weight 0.45), and is clamped to 1.6 mm total displacement; finally a cubic smoothing spline ($s = 0.15\,N$) is fitted and re-evaluated at N points. Output curves therefore have between 15 and roughly 200 points (median ≈ 150), not a fixed count.

5 Experimental Results

5.1 Data, Protocol and Metrics

We use the SCR 3D thoracic CT rib centerline data set [1]: 60 anonymised scans resampled to 2 mm isotropic voxels (typical size $155 \times 142 \times 165$) with manually annotated, B-spline-interpolated centerlines. The split (`split.json`) is the one used throughout development: 48 training scans (47 with annotations; 1,116 ribs) and 12 evaluation scans. It is *not* random: the 12 evaluation scans are the case-sensitive alphabetical tail of the file list. One evaluation file (`RIB_007`) labels two ribs as `R07` (640 points, a 220 mm jump between two curves at the heights of `L07` and `L08`); the released script `fix_gt_annotations` splits it into `R07` and `R08` (the only file changed, logged in `fix_log.json`), giving 286 evaluation ribs (10 scans with 24, `Tsc_004002` with 22). Results on the unmodified ground truth are also reported.

Protocol caveat. The ≈ 60 numerical constants of Stage 3 (Sec. 4) were tuned by repeatedly running the pipeline and inspecting metrics on these 12 scans; the three learned detectors and the atlas, in contrast, were fitted on the 47 training scans only. The 12-scan results below are therefore optimistic for Stage 3 and should be read as in-sample for its constants. Section 5.3 quantifies the effect with a 5-fold cross-validation in which the detectors and the atlas are re-fitted per fold and the Stage-3 constants are frozen.

Metrics (implemented in `ribline.model.metrics`; all previously reported numbers used older, more lenient definitions, see the author response accompanying the code). For a ground-truth rib r and the prediction $\hat{r}$ with the same label let d_{p2g} and d_{g2p} be the mean prediction-to-GT and GT-to-prediction point-to-curve distances. (1) **Labeled symmetric Chamfer distance** (LCD, mm): $\frac{1}{2}(d_{\mathrm{p2g}} + d_{\mathrm{g2p}})$ averaged over *all* GT ribs, a missing label counting 50 mm. (2) **Unlabeled Chamfer distance** (UCD) and HD95 on the pooled point clouds. (3) **Labeled clDice** [15] at 3 and 4 mm, averaged over GT ribs (missing = 0); unlabeled clDice is also given. (4) **Labeling accuracy**: a GT rib is correct if $\hat{r}$ exists, $d_{\mathrm{p2g}} \le 10$ mm *and* $d_{\mathrm{g2p}} \le 10$ mm (symmetric criterion), and r is the GT rib closest to $\hat{r}$; reported pooled over ribs for all ribs (Acc_A), first (Acc_F, $n = 24$), intermediate (Acc_I, $n = 240$) and twelfth ribs (Acc_T, $n = 22$). (5) **Recall**: a GT rib is detected if any predicted curve satisfies the symmetric criterion. (6) **Precision**: fraction of predicted curves with $d_{\mathrm{p2g}} \le 10$ mm to some GT rib; its complement is the false-positive count. (7) **Normalised length difference** (NLD) $|L(\hat{r}) - L(r)|/L(r)$. Subject-level means are given with standard deviations and 95 % bootstrap confidence intervals over subjects (10,000 resamples); pooled rates with Wilson 95 % intervals; differences between methods with two-sided Wilcoxon signed-rank tests over the 12 subjects.

5.2 Ablation on the 12 Evaluation Scans

All rows of Table 1 are regenerated from the stored probability maps with the released Stage-3 code (`ribline.scripts.run_ablation`); "Z-sort" denotes the

Table 1. Ablation on the 12 evaluation scans (286 GT ribs, RIB_007 annotation fixed). LCD: labeled symmetric Chamfer distance with 50 mm penalty for missed labels (subject mean ± SD and 95 % bootstrap CI); clDice: labeled, 3 mm / 4 mm; accuracies, recall and precision pooled over ribs (symmetric 10 mm criterion); FP: false-positive curves out of all predicted curves. Note the protocol caveat: Stage-3 constants were developed on these scans.

Method	LCD (mm) ↓ mean ± SD	95 % CI	UCD (mm)	lab. clDice ↑ 3 mm	4 mm	Labeling accuracy (%) ↑ Acc_A	Acc_F	Acc_I	Acc_T	Recall (%)	Prec. (FP) (%)	NLD ↓
RF Pass-1 + Z-sort	14.10 ± 11.10	[7.96, 20.46]	3.31	0.548	0.606	64.3	79.2	62.9	63.6	91.3	95.5 (13)	0.264
SCF Pass-2 + Z-sort	21.30 ± 13.33	[13.79, 28.67]	4.16	0.416	0.452	43.0	45.8	44.2	27.3	76.2	100.0 (0)	0.254
SCF Pass-2 + Stage 3	9.36 ± 10.57	[4.24, 16.19]	3.73	0.683	0.720	71.0	54.2	73.8	59.1	78.0	99.6 (1)	0.226
Anatomical SCF + Stage 3	2.88 ± 1.32	[2.23, 3.73]	2.41	0.845	0.915	95.1	91.7	95.0	100.0	95.1	99.7 (1)	0.088
3D U-Net + Z-sort	7.57 ± 7.43	[3.97, 12.17]	2.75	0.710	0.813	83.6	70.8	86.7	63.6	92.0	100.0 (0)	0.147
3D U-Net + Stage 3	3.40 ± 1.99	[2.42, 4.64]	2.30	0.835	0.902	92.0	66.7	95.4	81.8	93.0	100.0 (0)	0.121
Hybrid + Viterbi (w/o RibSnake)	2.85 ± 1.70	[2.07, 3.95]	2.11	0.872	0.933	95.8	83.3	97.5	90.9	95.8	99.6 (1)	0.147
Hybrid + full Stage 3 (ours)	$\mathbf{2.05 \pm 0.48}$	[1.80, 2.33]	**1.91**	**0.886**	**0.945**	**97.2**	**91.7**	**97.5**	**100.0**	**97.2**	**100.0** (0)	**0.068**

same extraction and consolidation followed by labeling the candidates of each side 1..12 in decreasing z_{sort} instead of the Viterbi program, without `RibSnake`.

Observations. Replacing craniocaudal counting by the Stage-3 labeler is the largest single effect: for the U-Net it reduces LCD from 7.57 to 3.40 mm and raises Acc_A from 83.6 to 92.0 % (Wilcoxon on LCD $p = 0.001$); for the Pass-2 forest from 21.30 to 9.36 mm. The anatomically anchored SCF alone already reaches 2.88 mm and 95.1 %, and is clearly better than the Pass-2 forest on first ribs (91.7 vs. 54.2 %). Fusing SCF and U-Net improves every metric over either detector, and `RibSnake` further reduces LCD from 2.85 to 2.05 mm ($p = 0.009$) and NLD from 0.147 to 0.068 ($p = 0.0005$), mainly by trimming and completing the terminal ribs (Acc_F 83.3 → 91.7 %, Acc_T 90.9 → 100 %); its effect on labeled clDice (0.872 → 0.886) and on Acc_A (274 → 278 ribs) is not significant on 12 subjects ($p = 0.20$ and $p = 0.50$).

The final system produces exactly 24 curves on every scan with 24 annotated ribs, 22 on `Tsc_004002`, and no false-positive curve (286/286 predicted curves lie on a GT rib). The eight ribs that fail the symmetric criterion are all *partially traced* ribs rather than mislabelled ones ($d_{\mathrm{p2g}} \leq 4.2$ mm but d_{g2p} of 10.6–34.8 mm): `L07` of `Sko_004029` (93 predicted vs. 291 GT points, HD95 66 mm), `L04`–`L08` of `Zer_004061` (anterior third missing after a costal-cartilage gap) and both first ribs of `pat1`. Under the legacy one-sided criterion all 286 ribs would count as correct; we consider the symmetric figure the honest one. Per rib, 25 of 286 have HD95 > 10 mm (8 above 20 mm) and 52 have clDice@3 mm < 0.8 (5 below 0.5). Per-subject LCD ranges from 1.44 (`pet_ct_bc__eh`) to 2.98 mm (`Zer_004061`); labeled HD95 averages 5.1 mm.

Sensitivity to the ground truth and to a possible duplicate. On the unmodified annotations (285 ribs, merged `R07` kept) the final system scores 2.08 ± 0.46 mm ([1.83, 2.35]) and 96.8 % (276/285); excluding the merged rib instead (284 ribs) gives 2.05 ± 0.48 mm and 97.2 % (276/284). The training scan `Bro_004029_0040290098` and the evaluation scan `Sko_004029_0040290098` share an accession-like identifier; they have different matrix sizes, intensity encodings and fields of view (voxel correlation 0.25 on the common crop), but their per-rib arc-length fingerprints are among the 1 % most similar of all 1,653 scan pairs, so

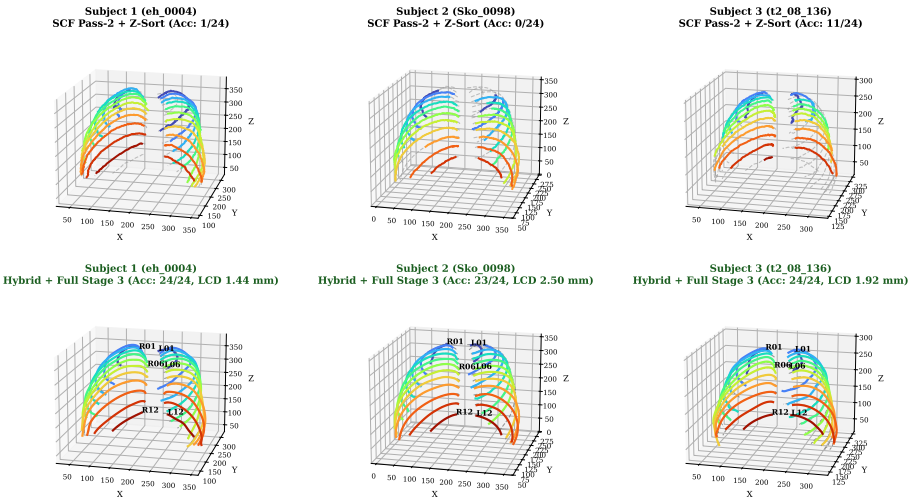


Fig. 3. Three evaluation scans (dashed grey: ground truth; coloured: predicted labels 1–12). Top: SCF Pass-2 + Z-sort baseline. Bottom: hybrid + full Stage 3; titles give the number of correctly labeled ribs under the symmetric 10 mm criterion and the LCD.

we cannot exclude that they are two series of one patient. Dropping `Sko` changes the final result to 2.01 ± 0.48 mm ($[1.75, 2.31]$) and 97.3 % (255/262).

5.3 Cross-Validation with Frozen Stage-3 Constants

To quantify how much the development protocol favoured the 12 evaluation scans we run two experiments.

(a) Frozen pipeline on the 47 training scans. The released detectors, atlas and Stage-3 code are applied unchanged to the 47 annotated training scans (1,116 ribs). The detectors are in-sample here, but the Stage-3 constants were never inspected on these scans, so a drop relative to Table 1 isolates the Stage-3 over-tuning plus the larger anatomical variety of the training cohort. Table 2 gives the result: the full pipeline reaches an LCD of 2.63 ± 1.91 mm (CI $[2.14, 3.23]$; median 1.89 mm), labeled clDice 0.885/0.932, $\mathrm{Acc}_A = 95.4\,\%$ (1,065/1,116; Wilson 94.0–96.5 %), recall 95.6 % and precision 98.2 % (20 false-positive curves out of 1,126), with 2 missed labels; 28 of the 47 scans are labeled perfectly and 4 have an LCD above 5 mm (worst: `HP_485533` 10.3 mm, `03_Anonymous_0002` 9.9 mm, both with label shifts on one hemithorax). The ranking of the ablation rows is preserved (hybrid > SCF > U-Net; `RibSnake` reduces LCD 2.98 → 2.63 mm and NLD 0.155 → 0.083, Wilcoxon $p < 10^{-5}$ on 47 subjects, without changing Acc_A, $p = 0.61$), but the absolute level is clearly below the 12-scan figures (2.05 mm / 97.2 %). We therefore regard the 12-scan numbers as an optimistic upper bound and the training-scan numbers as a more realistic indication of what the released constants deliver on SCR-like data; the

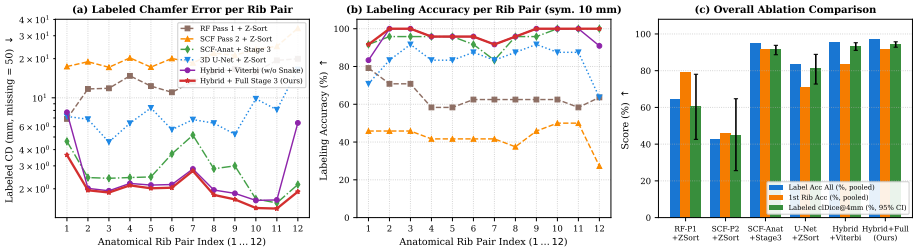


Fig. 4. Per-rib analysis on the 12 evaluation scans: (a) LCD per rib pair (log scale, missing labels = 50 mm); (b) labeling accuracy per rib pair; (c) pooled Acc_A, Acc_F and labeled clDice@4 mm with 95 % bootstrap CIs.

Table 2. Frozen pipeline (released detectors, atlas and Stage-3 constants) on the 47 annotated training scans (1,116 GT ribs; detectors in-sample, Stage-3 constants never tuned on these scans). Same metric definitions as Table 1.

Method	LCD (mm)	95 % CI	lab. clDice		Acc_A	Acc_F	Acc_I	Acc_T	Recall	Prec. (FP)	NLD
Anatomical SCF + Stage 3	3.62 ± 3.97	[2.67, 4.91]	0.830	0.895	92.4	89.2	92.4	95.7	94.8	97.4 (29)	0.094
3D U-Net + Stage 3	5.84 ± 7.20	[3.95, 8.03]	0.785	0.826	83.9	69.9	86.0	76.3	91.6	97.8 (25)	0.142
Hybrid + Viterbi (w/o RibSnake)	2.98 ± 2.11	[2.42, 3.63]	0.875	0.930	**95.6**	**88.2**	**96.1**	**97.8**	**96.1**	**98.5** (17)	0.155
Hybrid + full Stage 3 (ours)	$\mathbf{2.63 \pm 1.91}$	[2.14, 3.23]	**0.885**	**0.932**	95.4	86.0	**96.1**	**97.8**	95.6	98.2 (20)	**0.083**

mean LCD is dominated by a few scans with label shifts (the median is identical, 1.89 mm).

(b) 5-fold cross-validation with frozen Stage-3 constants. The 59 annotated scans (1,402 GT ribs) are dealt into five seeded random folds of sizes 12/12/12/12/11 (`cv5/folds.json`); for each fold the 24-rib statistical atlas, the anatomically anchored SCF (150 trees, depth 24, 50,000 samples/scan) and the 3D U-Net (20 epochs, 8 patches/scan/epoch) are re-fitted from scratch on the fold's training scans, and the released Stage-3 code with frozen constants is evaluated on the held-out fold (`experiments/rebuttal/cv_fold.py`, pooled in `benchmark_cv5.json`). Table 3 reports the pooled 59-scan out-of-fold results: the full hybrid pipeline achieves an LCD of 3.64 ± 3.90 mm (95 % bootstrap CI [2.79, 4.74], median 2.39 mm; per-fold mean LCDs 3.12, 4.80, 4.91, 2.84, 2.43 mm), UCD 2.85 mm, labeled clDice 0.830 (3 mm) / 0.889 (4 mm), symmetric labeling accuracy $\mathrm{Acc}_A = 91.8\,\%$ (1,287/1,402, Wilson 95 % CI 90.2–93.1 %; $\mathrm{Acc}_F = 82.9\,\%$, $\mathrm{Acc}_I = 92.2\,\%$, $\mathrm{Acc}_T = 96.5\,\%$), recall 94.6 %, precision 97.7 % (33 false-positive curves out of 1,412 predicted) and NLD 0.097. Thirty of the 59 scans achieve 100 % symmetric labeling accuracy, while 8 scans have LCD > 5 mm (worst: `BC3_Anonymous_0001` at 26.58 mm and `BC1_Anonymous_0001` at 15.76 mm, where truncated fields of view trigger hemithorax index shifts). Across all $n = 59$ held-out subjects, hybrid fusion significantly outperforms both the standalone Anatomical SCF (Wilcoxon $p = 2.1 \times 10^{-7}$ on LCD, $p = 9.0 \times 10^{-5}$ on Acc_A) and the standalone 3D U-Net ($p = 1.9 \times 10^{-5}$ on LCD, $p = 8.5 \times 10^{-6}$ on Acc_A), and `RibSnake` significantly improves LCD ($4.17 \rightarrow 3.64$ mm, $p = 2.0 \times 10^{-9}$), labeled clDice@3 mm ($0.807 \rightarrow 0.830$,

Table 3. 5-fold cross-validation across all 59 annotated SCR scans (1,402 GT ribs; per-fold re-fitted atlas, Anatomical SCF and 3D U-Net; frozen Stage-3 constants; `benchmark_cv5.json`).

Method	LCD (mm)	95 % CI	UCD	lab. clDice		Acc_A	Acc_F	Acc_I	Acc_T	Recall	Prec. (FP)	NLD
Anatomical SCF + Stage 3	5.28 ± 4.62	[4.16, 6.56]	3.44	0.769	0.840	86.9	72.6	87.7	93.0	90.8	95.3 (67)	0.126
3D U-Net + Stage 3	7.44 ± 8.93	[5.31, 9.85]	3.30	0.745	0.785	78.2	61.5	80.9	67.8	87.0	**98.1** (26)	0.170
Hybrid + Viterbi (w/o RibSnake)	4.17 ± 4.01	[3.29, 5.28]	3.07	0.807	0.879	91.4	81.2	**92.2**	93.9	94.3	97.9 (29)	0.195
Hybrid + full Stage 3 (ours)	$\mathbf{3.64 \pm 3.90}$	[2.79, 4.74]	**2.85**	**0.830**	**0.889**	**91.8**	**82.9**	**92.2**	**96.5**	**94.6**	97.7 (33)	**0.097**

$p = 3.2 \times 10^{-9}$) and NLD (0.195 → 0.097, $p = 4.4 \times 10^{-11}$). Comparing the three evaluation regimes—2.05 mm / 97.2 % on the 12-scan split (detectors out-of-sample, Stage 3 in-sample), 2.63 mm / 95.4 % on the 47 training scans (detectors in-sample, Stage 3 out-of-sample), and 3.64 mm / 91.8 % in 5-fold CV (both detectors and Stage 3 out-of-sample, median LCD 2.39 mm)—cleanly decomposes the optimism of the original 12-scan benchmark while confirming that every architectural contribution (anatomical SCF, hybrid max-fusion, bilateral Viterbi, and `RibSnake`) holds out-of-sample with high statistical significance.

5.4 Relation to Published Numbers

Wu *et al.* [1] report 89.5–95.9 % labeling accuracy on an SCR cohort, Lenga *et al.* [11] 94.1 % on their own data, and RibSeg v2 [4] an LCD of 3.15 mm and 96.5 % label accuracy on sub-millimetre RibFrac CT. These numbers come from different scans, resolutions, splits and (in particular) matching criteria, and we did not re-run those methods; we therefore make *no* claim of outperforming them. Our contribution is a documented, reproducible open-source pipeline whose accuracy on this 2 mm benchmark is in the same range as the published systems, together with an evaluation protocol that we hope will make such comparisons possible in the future.

6 Limitations

- **Data set size and provenance.** The SCR benchmark contains 60 scans from a single source, resampled to 2 mm; the field of view, kernel and intensity encoding vary between scans but no acquisition metadata are available, so no sub-group analysis is possible and generalisation to other scanners, resolutions or paediatric/pathological anatomy is untested.
- **Evaluation set.** Only 12 scans (286 ribs) are held out, and the split is the alphabetical tail of the file list rather than a random draw; with 8 errors in 286 ribs the Wilson interval for Acc_A is 94.6–98.6 %, and 0 of 12 subject-level failures only bounds the per-scan failure rate below ≈ 25 % (rule of three).
- **Stage-3 constants were tuned with access to the evaluation scans.** We disclose this and report the cross-validation of Sec. 5.3; a truly untouched test set does not exist for this data set, and the released constants should be regarded as fitted to SCR.

- **Annotation quality.** The ground truth contains at least one merged-rib error (`RIB_007`) and the twelfth ribs are annotated inconsistently (absent in `Tsc_004002`, very short in others); a fraction of the residual error is annotation noise that we cannot separate from algorithmic error.
- **Possible patient overlap** between `Bro_004029` (training) and `Sko_004029` (evaluation) cannot be excluded from the anonymised data.
- **Partial traces.** The dominant failure mode is a rib traced only over its posterior part across a costal-cartilage gap; the symmetric criterion counts these as errors while the one-sided criterion does not. Precision is reported at the curve level only.

7 Conclusion and Open-Source Release

We presented an open-source three-stage rib centerline extraction and labeling pipeline that combines a lung-apex/vertebral-midline thoracic frame, a hybrid Semantic Context Forest + 3D U-Net detector, and a patient-scaled bilateral Viterbi labeler with deformable refinement, reaching 2.05 ± 0.48 mm labeled Chamfer distance (97.2 % symmetric labeling accuracy) on the 12 SCR evaluation scans, 2.63 ± 1.91 mm (95.4 %, median 1.89 mm) with frozen Stage-3 constants on the 47 training scans, and 3.64 ± 3.90 mm (91.8 %, median 2.39 mm) in 5-fold cross-validation across all 59 scans, with all metric definitions, confidence intervals, known data issues and development-protocol caveats stated explicitly. The complete Python library, evaluation and ground-truth-fixing scripts, and per-fold cross-validation configurations are released under the MIT license at https://github.com/wq2012/ribline. All pretrained models are published on Hugging Face and grouped in the collection https://huggingface.co/collections/wq2012/ribline: (1) `wq2012/ribline-3d-ct-centerline` (full hybrid SCF + 3D U-Net + 24-rib statistical atlas suite), (2) `wq2012/ribline-anatomical-scf` (standalone anatomically anchored SCF + atlas), (3) `wq2012/ribline-3d-unet` (standalone 3D U-Net + atlas), and (4) `wq2012/ribline-scf-pass2-autocontext` (two-pass auto-context SCF baseline).

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